

Officiating Partnership



Development by

Liv5 Solutions

In Collaboration with



Liv5Studio

Subject of



Project Liv5-IRS

Final Report

Confidential

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Executive Summary

Project Overview

In recent years, basketball has grown rapidly in Vietnam, with a large number of school, university, and semi-professional tournaments. However, most of these tournaments still cannot access professional Instant Replay Systems because of high cost and complex deployment requirements. Current livestreaming software is designed primarily for broadcasting and does not adequately meet the need to support referees in reviewing situations.

To address this problem, Liv5Solutions has been developing Liv5-IRS (the Liv5 Instant Replay System) — a solution that helps basketball referees review game situations. The system provides features such as multi-angle replay, slow motion, frame-by-frame review, synchronization of up to 16 cameras, and integrated AI to support player tracking, image enhancement, and situation search. AI plays a supporting role only; the final decision remains with the referee in accordance with FIBA rules.

During development, the team conducted market research, user surveys, and competitor analysis. The results show that groups such as referees, organizers, and athletes all recognize a strong need to apply technology in order to improve the transparency and fairness of matches.

Beyond theoretical research, Liv5-IRS was successfully demonstrated at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026). The system received positive feedback from the leadership of the Vietnam Basketball Federation (VBF), the referee team, and the Head of the VBF Referees Committee. In addition, several sponsors present at the tournament expressed a desire to cooperate with Liv5Solutions on future projects. Liv5-IRS is currently working with Minh Khang Center to integrate the software into an Instant Replay System solution package for domestic basketball tournaments.

Liv5Solutions' business model focuses on licensing the software on a per-tournament basis through cooperation with livestreaming providers or organizers, while partnering with hardware providers to reduce deployment costs for customers.

With reasonable cost, flexible deployment, short training time, and scalability, Liv5-IRS aims to become a widely adopted Instant Replay System for student and semi-professional basketball tournaments in Vietnam, and to gradually expand into professional tournaments and the regional market.

Chapter 1. Business Opportunity

1.1. Industry and Market Context

In recent years, sport in general and basketball in particular have grown strongly in Vietnam. Alongside professional tournament systems such as the VBA (Vietnam Basketball Association), the number of school, university, corporate, and community tournaments has also risen quickly. Many tournaments — such as the National Student Basketball Championship, the Vietnam Collegiate Basketball Championship (VCBC), the GMB League, and various 3x3 events — are held annually, helping to expand the player community and attract the attention of businesses and audiences. According to a VBA representative, community basketball in Vietnam is sustaining growth of roughly 20–30% per year, especially among students and young people. This growth drives demand for higher-quality tournament organization and refereeing.

In parallel, the trend of applying technology to sport (Sports Technology) is becoming increasingly common. Worldwide, video-assisted officiating systems such as VAR in football, Hawk-Eye in tennis, and the Instant Replay System (IRS) in basketball have become standard at many professional tournaments. For basketball, FIBA has issued procedures for using the Instant Replay System to help referees review key situations and make more accurate and transparent decisions.

However, in Vietnam, adoption of the Instant Replay System is still limited. Most school, university, community, and semi-professional tournaments lack the resources to invest in professional systems because of the high cost, the requirement for dedicated hardware, and the need for a technical operations team. Meanwhile, live broadcasting software such as OBS Studio or vMix is designed mainly for program production and livestreaming, not for the referee's review workflow. As a result, many tournaments still rely entirely on direct observation, especially in fast-moving situations that have a major impact on the outcome of the match.

This reality creates an opportunity for Instant Replay solutions that are affordable and easy to deploy while still meeting referees' needs. Rather than targeting international tournaments with expensive hardware systems, a software solution for student and semi-professional tournaments is better suited to real conditions in Vietnam.

In this context, Liv5-IRS was developed as a software-based Instant Replay System that can be deployed on existing hardware or through equipment-providing partners. The system focuses on referees' needs, with functions such as multi-angle replay, slow motion, frame-by-frame review, and multi-camera synchronization, while integrating AI technologies to support situation analysis without replacing the referee's decision-making authority.

1.1.1. Overview of the Liv5 Instant Replay System (Liv5-IRS)

The Liv5 Instant Replay System (Liv5-IRS) is a system that helps referees review in-game situations through video recorded from multiple camera angles. Under FIBA rules, the IRS does not replace the referee's role but acts as a support tool for situations that are permitted to be reviewed under the Official Basketball Rules. Referees can use the system to rewind, view in slow motion, or examine frame by frame in order to more accurately assess contested situations before making a final decision.

Professional IRS systems today are usually deployed at international or high-level tournaments — a leading example being EVS Xeebra. These systems can synchronize multiple cameras, provide high-quality images, and meet the strict standards of international sports federations. However, the large



investment cost and the technical infrastructure requirements make these solutions unsuitable for most tournaments in Vietnam.

Meanwhile, live broadcasting software such as OBS Studio or vMix can review video but is built for program production and broadcast control. The interface, operating workflow, and functions of this software are not optimized for referees, making it difficult and inefficient to use in a competition environment.

Liv5-IRS is positioned as an Instant Replay System aimed at student and semi-professional tournaments, focused on closing the gap between very expensive professional IRS systems and general-purpose livestreaming software that does not meet the needs of match officiating.

1.1.2. Market Size

Although there is no public database that fully records the number of basketball tournaments held each year in Vietnam, information gathered from tournament organizers and industry partners indicates that the number of tournaments is trending upward year over year. Because of commercial sensitivity, many figures related to tournament scale and organizing revenue are not widely published. The market-size assessments in this report are therefore built on a combination of publicly available data and information provided by industry partners, for analytical purposes.

According to statistics compiled by Liv5Studio — a media company that regularly works with basketball tournament organizers in Hanoi — as of July 2026, Vietnam had held 4 national-level tournaments, comprising 2 organized by the Vietnam Basketball Association (VBA) and 2 organized by the Vietnam Basketball Federation (VBF). In Hanoi alone, there were 2 city-level tournaments and 1 district-level tournament organized by an inter-level (multi-grade) school in the area. Thus, within only the tournaments tracked and compiled by Liv5Studio, 7 tournaments were held in the first half of 2026. Because these figures do not include many grassroots, student, and other tournaments held across the country, the true size of the market is assessed to be considerably larger.

In addition to the professional tournament system, Vietnam has dozens of school, university, corporate, and community tournaments held annually in Hanoi, Ho Chi Minh City, and many other provinces and cities. Tournaments such as the Vietnam Collegiate Basketball Championship (VCBC), the GMB League, the National Student Basketball Championship, national summer tournaments, and many 3x3 events demonstrate the strong growth of the basketball movement. This is precisely the target customer segment Liv5-IRS is aiming for in its initial phase.

Alongside the increase in the number of tournaments, organizers are paying more and more attention to raising professional quality, ensuring transparency in officiating, and improving the audience experience. This trend creates favorable conditions for applying technology solutions such as the Instant Replay System to support referees, enhance organizing quality, and increase the professionalism of tournaments in the future.

1.2. Market Problems

The rapid growth of basketball in Vietnam has driven increasing demand for higher-quality tournament organization and officiating. However, the ability to apply technology to support referees has not kept pace with the market. For most school, university, semi-professional, and community tournaments, deploying a professional Instant Replay System (IRS) still faces significant barriers in cost, technical infrastructure, and operating staff.

1.2.1. The cost of deploying a professional Instant Replay system is still too high

Today, the Instant Replay systems used at international tournaments are mostly integrated software-and-dedicated-hardware solutions that require synchronized technical infrastructure and a specialized operating team. A complete system typically includes multiple high-quality cameras, a processing server, signal-synchronization equipment, and a dedicated control desk, pushing the total investment and deployment cost beyond the financial reach of most tournaments in Vietnam.

For many organizers, especially student and semi-professional tournaments, the budget is usually prioritized for the venue, media, prizes, and organizing activities. Meanwhile, the number of matches in each tournament is not large enough to justify the investment in a professional Instant Replay system. As a result, although they recognize the benefits of this technology, many organizations still do not have the means to access and deploy it.

1.2.2. Lack of an Instant Replay solution suited to the real conditions of tournaments in Vietnam

Currently, basketball tournaments in Vietnam mainly have access to two groups of situation-review solutions. The first group is professional Instant Replay systems used at international tournaments. These systems provide the full set of referee-support functions, such as viewing multiple angles simultaneously, slow motion, frame-by-frame viewing, and near-instant replay. However, the very high investment cost, the requirement for dedicated hardware, and the need for a specialized technical team make it difficult for most student, semi-professional, and local tournaments to deploy them.

The second group is common livestreaming and recording software such as OBS Studio or vMix. Although they have the advantages of low cost, easy deployment, and wide use for live broadcasting, this software is designed for program production rather than for supporting referees in match officiating. Its interface and workflow focus on managing feeds, switching scenes, and controlling the program, whereas referees need fast access to a situation that just occurred, the ability to view multiple angles simultaneously, slow motion, or frame-by-frame viewing to reach a decision within a short time. Having to operate non-specialized software or depend on a technician lengthens processing time and reduces the referee's autonomy.

The difference between these two groups has created a clear market gap. One side provides the full set of features but at high cost and with difficult deployment, while the other is low-cost but does not properly meet the referee-support workflow. This shows a need for an Instant Replay system that is affordable, easy to deploy, optimized for referees, and suited to the real conditions of basketball tournaments in Vietnam.

1.2.3. Controversial decisions harm tournament reputation

In basketball, many situations — such as out-of-bounds calls, contact fouls, the moment the ball leaves the hand at the buzzer, or determining which player last touched the ball — happen very quickly and can directly affect the outcome of the match.

Without a video-review support tool, referees are forced to decide based on direct observation. This increases the likelihood of disputes among athletes, coaches, and spectators, especially in decisive situations.



Moreover, in an era of powerful social media, controversial situations can spread quickly, affecting the reputation of the organizer, the referee team, and the image of the tournament. For tournaments with sponsors or live broadcasting, this also affects brand value and the ability to attract future partners.

1.2.4. Business opportunity for Liv5Solutions

The market gap above shows that the need for an Instant Replay system that is affordable, easy to deploy, and designed specifically for referees is well-founded. This is a need not only of professional tournaments but also, increasingly clearly, of student and semi-professional tournaments as organizing quality continues to rise.

Liv5-IRS was developed to meet that need through a software-based Instant Replay solution that can be deployed on existing infrastructure or combined with equipment-providing partners to significantly reduce investment cost. The solution aims to support referees during situation review, while ensuring that the final decision remains with the referee in accordance with FIBA rules.

1.3. Target Customers

Identifying the right target customers plays an important role in product development and in shaping Liv5Solutions' commercialization strategy. Based on survey results, discussions with referees, tournament organizers, and basketball experts, and real-world testing, the team has determined that Liv5-IRS does not target basketball players or spectators as direct customers, but instead focuses on the organizations responsible for organizing and running tournaments.

Although fans and athletes benefit from greater match transparency, the decision to invest in and deploy an Instant Replay System rests mainly with organizers and federations. Liv5-IRS therefore adopts a Business-to-Business (B2B) model as its main approach in the initial phase.

1.3.1. Primary Customer Segment

The most important target customers for Liv5-IRS are the organizers of student and semi-professional basketball tournaments and the units responsible for refereeing (that is, the federation side — VBF).

This is the group directly responsible for the professional quality, officiating, and image of the tournament. Any dispute over a refereeing decision directly affects the organizer's reputation, so they have a strong incentive to invest in solutions that improve the fairness and transparency of matches.

Characteristics of this customer group include:

- Having a tournament budget, but one constrained by cost.
- Being interested in improving the quality of the match.
- Needing a solution that can be used immediately with the existing referee team.
- Wanting fast deployment, easy operation, and no need to invest in expensive hardware.

This is also the customer group Liv5-IRS prioritizes reaching through direct cooperation with tournament organizers.

1.3.2. Secondary Customer Segment

The secondary customer group comprises livestreaming, recording, and sports media providers.

In recent years, many basketball tournaments in Vietnam have hired livestreaming providers to record and broadcast live on platforms such as Facebook or YouTube. These providers have typically already invested in camera systems, recording equipment, and technical staff for content production.



Rather than investing in and operating its own recording system, Liv5-IRS can coordinate with livestreaming providers to leverage their existing camera infrastructure and video feeds, and thereby deploy the Instant Replay function for referees. This cooperation model reduces deployment costs for organizers while creating additional business opportunities for livestreaming providers through offering Instant Replay support services at tournaments.

Under the proposed business model, Liv5-IRS cooperates with livestreaming providers by supplying the software together with deployment services. The cost is charged per tournament or according to the organizer's needs, depending on the scale, duration, and number of cameras used.

1.3.3. Strategic Customer Segment

Besides tournament organizers, federation-affiliated units and pioneering suppliers of basketball equipment and materials are strategic customers for Liv5Solutions.

According to a draft currently under consideration, the Vietnam Basketball Federation (VBF) is expected to update certain provisions of Vietnam's basketball rules, potentially including the certification of systems and software such as the upcoming Liv5-IRS. A proposed rule encouraging wider use of an IRS at tournaments is also reportedly under discussion.

The Vietnam Basketball Federation's agreement to allow Liv5Solutions to demonstrate the product at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026), and the positive feedback from leaders at all levels and from the match officiating referee team, is an important validation step that helps establish a foundation for the system to be considered for wider use in more tournaments in the future.

Currently, Liv5Solutions has received the Vietnam Basketball Federation's approval to pilot Liv5-IRS across entire tournaments, including national and local tournaments. The VBF is monitoring the progress of completing Liv5-IRS so that it can grant certification quickly and put the system into direct operation as soon as possible.

In addition, Liv5Solutions cooperates with Minh Khang Center — a supplier specializing in basketball court materials — so that Vietnam will soon have its first licensed IRS system used at all tournaments. Minh Khang Center will be responsible for importing the Liv5-IRS hardware, while Liv5Solutions' role is to provide the software that runs on that system.

1.3.4. Customer Analysis Using the STP Model

The STP model (Segmentation – Targeting – Positioning) is used to identify target customers and build a suitable market-approach strategy. Within it, Segmentation divides the market into groups of customers with similar characteristics; Targeting identifies the customer group the business prioritizes serving; and Positioning positions the product in the market to differentiate it from existing solutions.

At this stage, the team performs market segmentation based on four criteria — Demographic, Geographic, Psychographic, and Behavioral — as a basis for selecting target customers and building a positioning strategy for the Instant Replay System (Liv5-IRS).

Demographic

Target customers are mainly organizations, businesses, and units operating in the sports field, including tournament organizers, sports centers, schools, universities, livestreaming providers, and basketball federations. Decision-makers are typically the head of the organizing committee, the head of the



technical/professional department, a federation representative, or the person responsible for running the tournament.

Geographic

In the initial phase, Liv5-IRS focuses on deployment in Hanoi and Ho Chi Minh City, where most large-scale basketball tournaments are concentrated and the basketball ecosystem is well-developed. Once the product is refined and the partner network is expanded, deployment will extend to other provinces and cities nationwide.

Psychographic

Target customers are organizations that want to raise the quality of tournament organization, build a professional image, and earn the trust of athletes, spectators, and sponsors. They are willing to invest in technology solutions if these deliver real effectiveness, are easy to deploy, and fit the budget.

Behavioral

Customers prioritize solutions that can be deployed quickly, are easy to use, and do not require large infrastructure investment. They value technical support, after-sales service, and the product's stability during the tournament. Customers who have previously experienced refereeing disputes, or who frequently organize tournaments, tend to be more interested in an Instant Replay System.

Chapter 2. Market and Customer Validation

2.1. Competitor Analysis

One important step in developing Liv5-IRS was analyzing existing market solutions in order to identify the market gap and build an appropriate competitive advantage. Through this research, the team found that current solutions can be divided into two main groups: direct competitors and indirect competitors.

Direct competitors are Instant Replay Systems designed specifically for referees at sports tournaments. Indirect competitors, meanwhile, include livestreaming or recording software that can support video review but was not developed for the purpose of match officiating.

Analyzing both groups helps the team clearly understand Liv5-IRS's position in the market and identify the key differentiators to focus on during product development.

2.1.1. Direct Competitors

Currently, the Vietnamese market has no Instant Replay System developed and commercialized with the same orientation of serving student and semi-professional basketball tournaments. Therefore, within the scope of this research, Liv5Solutions selects EVS Xeebra — an Instant Replay system widely used at international tournaments — as a reference point for analyzing and comparing technology, features, and deployment models. This ensures the product is assessed on an objective basis consistent with industry standards.

EVS Xeebra

EVS Xeebra is one of the Instant Replay Systems widely used at professional basketball tournaments and international tournaments organized by FIBA. Developed by EVS Broadcast Equipment and designed specifically for referees, it allows multi-angle video review with low latency and high image quality.

Notable functions of EVS Xeebra include:

- Simultaneous multi-camera synchronization.
- Real-time review.
- Slow motion and frame-by-frame viewing.
- An interface optimized for referees.
- High stability in a professional competition environment.

However, EVS Xeebra is aimed mainly at international and professional tournaments. Deploying the system requires dedicated hardware, a large investment, and an experienced technical team. This is a significant barrier for student and semi-professional tournaments in Vietnam.

Although EVS Xeebra is regarded as the standard in the Instant Replay field, the team does not view it as a direct competitor in terms of customer segment. Instead, EVS Xeebra serves as a technological reference for Liv5-IRS during development, while Liv5-IRS chooses a direction better suited to the real conditions of the Vietnamese market.

Unlike EVS Xeebra, Liv5-IRS is developed with a software-first approach, leveraging the existing infrastructure of organizers or livestreaming partners instead of requiring investment in a dedicated hardware system.

The product focuses on three main factors:

- Deployment cost suited to domestic tournaments.
- A simple interface that referees can use after about five minutes of instruction.
- The ability to integrate AI to support match officiating without replacing the referee's role.

The system currently supports synchronizing up to 16 cameras and can be deployed on a range of hardware configurations. On high-specification computers, Liv5-IRS provides full AI functions such as player tracking, image enhancement, and an AI assistant. On general-purpose devices, the system can still be used with its core functions, omitting only certain AI features that require large processing resources.

2.1.2. Indirect Competitors

Besides dedicated Instant Replay systems, many tournaments in Vietnam currently use livestreaming software such as OBS Studio or vMix to record and broadcast live.

OBS Studio

OBS Studio is open-source software widely used in livestreaming. It supports managing multiple video sources, broadcasting live to social platforms, and high-quality recording.

The greatest advantage of OBS Studio is that it is free, easy to access, and backed by a large user community. However, OBS is not designed for refereeing work. Reviewing a situation takes many steps, there is no interface optimized for decision-making, and it does not support specialized functions such as angle synchronization or fast, FIBA-workflow control.

vMix

vMix is commercial livestreaming software used by many professional media units. Compared with OBS Studio, vMix has more advanced features such as Instant Replay, Slow Motion, and multi-camera management.

However, vMix's main purpose is still television program production and livestreaming, not referee support. Using the software still requires an experienced technician and does not fully meet the officiating workflow of a basketball tournament.

2.1.3. Competitor Comparison Summary

Criterion	EVS Xeebra	OBS Studio	vMix	Liv5-IRS
Primary users	Professional referees	Livestreaming	Livestreaming	Referees
Main purpose	Instant Replay	Broadcasting	Broadcasting	Instant Replay
Multi-camera support	Yes	Yes	Yes	Yes (up to 16 cameras)
Slow motion	Yes	Yes	Yes	Yes
Frame-by-frame	Yes	Limited	Yes	Yes
AI support	No	No	No	Yes
Referee-optimized interface	Yes	No	No	Yes

Deployment cost	Very high	Low	Medium	Affordable
Fit for Vietnamese tournaments	Limited	Medium	Medium	High
Certification	FIBA Certified	No	No	VBF certification pending

Table 2.1. Comparison of Liv5-IRS with existing solutions

2.1.4. Liv5-IRS's Competitive Advantage

Through this analysis, the team found that Liv5-IRS does not compete directly with EVS Xeebra on technology in the international-tournament segment. Instead, the project focuses on the needs of student and semi-professional tournaments — a segment that still lacks solutions appropriate in both cost and deployability.

Liv5-IRS's competitive advantage comes from combining reasonable cost, flexible deployment, and a simple operating workflow. Rather than requiring a dedicated technical team, the system is designed so that referees can operate it directly after about five minutes of training. This significantly reduces staffing costs and preparation time before each tournament.

In addition, Liv5-IRS is developed with a direction of integrating artificial intelligence as a support tool, improving referees' effectiveness while ensuring that the final decision belongs to a human, fully in keeping with the spirit of the FIBA Basketball Rules. This is a differentiated direction compared with many AI solutions that aim to automate the decision-making process.

Beyond technology, the fact that Liv5-IRS has been tested at NYCC 3x3 2026, received positive feedback from the leadership of the Vietnam Basketball Federation (VBF) and the Head of the VBF Referees Committee, and is cooperating with Minh Khang Center, are important advantages that help demonstrate the product's feasibility under real conditions.

2.2. Survey and Customer Validation

2.2.1. Research Objectives

After identifying the business opportunity and market gap, the team conducted customer research to verify the initial assumptions about demand for an Instant Replay System in Vietnam. The research aimed to assess customers' level of interest in the solution, identify the problems customers face in organizing and running tournaments, and gather input to refine the product before real-world testing.

Beyond verifying the feasibility of the idea, the research results were also used to adjust the Value Proposition Canvas and the Business Model Canvas, and to build a commercialization strategy suited to market needs.

2.2.2. Research Methods

The team combined two research methods — quantitative and qualitative — to ensure that the data collected was both representative and reflective of customers' real-world perspectives.

For quantitative research, the team built an online questionnaire and collected 56 valid survey responses from people involved in organizing and participating in basketball tournaments in Vietnam. Respondents included referees, athletes, coaches, tournament organizers, livestreaming providers, and

basketball fans. The survey focused on four main areas: demand for using an Instant Replay System, willingness to pay, functions of interest, and expectations for a referee-support system.

In parallel with the survey, the team held direct discussions with referees, organizers, and people experienced in tournament organization to gain deeper insight into officiating workflows, real-world difficulties, and expectations for an Instant Replay system suited to conditions in Vietnam.

After completing the research phase, the team proceeded to test the product at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026). This served as field validation, helping the team assess the system's operability in a competition environment and gather direct feedback from stakeholders.

2.2.3. Survey Sample Information

In total, the team collected 56 valid responses.

Survey participants included various groups within the basketball ecosystem to ensure a multi-dimensional perspective, including:

- Basketball referees.
- Tournament organizers.
- Coaches.
- Athletes.
- Livestreaming providers.
- Basketball fans.

Choosing multiple groups allowed the team not only to assess the needs of those who directly use the system but also to understand the expectations of those affected by refereeing decisions, thereby building a solution better suited to the entire basketball ecosystem.

Bạn thuộc nhóm dưới đây nào?
56 câu trả lời



Figure 2.1. *Composition of survey respondents.*

2.2.4. Survey Results

Demand for applying an Instant Replay System

The survey results show that most participants regard an Instant Replay System as a necessary technology for domestic basketball tournaments. Many respondents believe that applying the technology will help increase transparency in refereeing decisions, reduce disputes between teams, and raise the professionalism of tournaments.

In particular, referees and organizers showed the highest level of interest, as they are the parties directly responsible for the professional quality of matches.

Willingness to use

One important survey result is that most participants said they are willing to use or recommend using an Instant Replay System if the solution is affordable and simple to operate.

This shows that the market's biggest barrier is not awareness of the product's value but mainly deployment cost and access to the technology.

This is precisely the core assumption Liv5-IRS aims to address through a software model with cost significantly lower than traditional Instant Replay systems.

Functions of interest

Through the survey, the team found that customers do not expect a system with too many complex functions but instead prioritize features that directly serve match officiating.

The functions rated most important include:

- Multi-angle replay.
- Slow motion.
- Frame-by-frame viewing.
- Fast situation search.
- A simple interface.
- Fast response time.

In addition, many participants expressed interest in applying AI to support referees. However, most believe that AI should only support analysis rather than make decisions on the referee's behalf.

This result fully aligns with Liv5-IRS's development direction, in which AI is used to track players, enhance image quality, and support situation search, while the final decision remains with the referee in accordance with FIBA rules.

Đánh giá mức độ hữu ích của các tính năng sau (1-5)

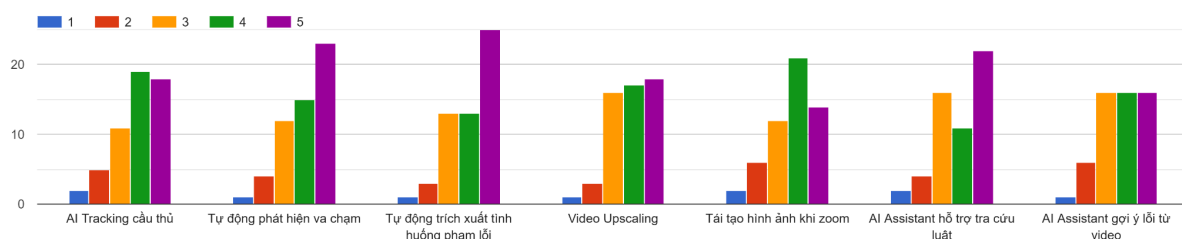


Figure 2.2. Functions of interest to customers.

2.2.5. Customer Insights

After the survey, interviews, and discussions with stakeholders, the team found that customers' needs do not stop at owning a situation-review system; more importantly, they want a solution suited to the conditions of organizing tournaments in Vietnam. Customers value products that can solve real problems at a reasonable cost, are easy to deploy, and do not change their existing operating workflow.

For tournament organizers, the most valued factor is the ability to improve fairness and transparency in refereeing decisions. They expect that applying technology will help reduce disputes, raise the tournament's professional image, and build trust with teams, spectators, and sponsors.

Referees, meanwhile, prioritize a system that can support fast decision-making during the match itself. Rather than many complex features, they want an intuitive interface, simple operation, and instant situation retrieval that does not interrupt the pace of play.

For livestreaming providers, the team found that they are willing to cooperate if the solution can leverage their existing camera systems and infrastructure. Adding the Instant Replay function not only improves service quality but also opens opportunities to reach tournaments that need referee-support technology.

From these results, the team determined that the core value customers want lies not in the number of features but in the ability to balance effective referee support, deployment cost, and ease of use. This is also the basis for refining the Value Proposition Canvas and guiding product development in subsequent phases.

2.2.6. Marketing Insights (4P)

Product

Survey results and discussions with organizers, referees, and operators show that customers do not expect an Instant Replay system with too many complex functions for television program production. Instead, what they care about most is the ability to help referees make fast, accurate, and convenient decisions in situations that require review.

Functions such as instant replay, slow motion, frame-by-frame viewing, multi-angle synchronization, and fast situation retrieval are seen as more valuable than editing- or broadcast-oriented features. Users also want a system with an intuitive, easy-to-use interface that does not require much training time.

From these results, the team determined that the core value of Liv5-IRS is to support referees rather than to support program production. The product is therefore developed to optimize the operating workflow, reduce the number of steps needed to review a situation, and integrate AI modules such as AI Tracking, AI Image Enhancement, and AI Assistant to support analysis. The AI features play a supporting role only and do not replace the referee's decision, consistent with the rules and spirit of the FIBA Basketball Rules.

Price

Through surveys and interviews with target customers, the team found that most tournaments in Vietnam — especially student, semi-professional, and local tournaments — have relatively limited organizing budgets. Investing in a professional, high-cost Instant Replay system exceeds the financial capacity of many organizers.

Rather than owning their own system, customers want to use the service per tournament or per deployment to optimize the initial investment. Customers are also willing to pay if the product delivers clear value in improving officiating quality and reducing disputes in key situations.

This insight is the basis for Liv5-IRS's choice of a Value-based Pricing strategy combined with a flexible service model. Specifically, the project offers two main forms of cooperation: providing a software package to livestreaming providers at a price of VND 5–10 million per tournament lasting about four weeks, or cooperating directly with tournament organizers at a starting price of VND 5 million per tournament, depending on scale and deployment requirements.

Place

One notable finding from the research is that customers do not want to invest in additional dedicated infrastructure solely to use Instant Replay in only a few tournaments per year. What they want is a solution that can be deployed quickly on existing infrastructure without significantly changing the tournament organizing workflow.

Based on that need, Liv5-IRS was developed to be compatible with many types of cameras and computers, allowing customers or livestreaming providers to make full use of their existing recording systems. For tournaments that lack sufficient equipment, the system can also be deployed through partners providing equipment and technical services.

In addition to supplying directly to organizers, Liv5-IRS also expands market access through livestreaming providers and strategic partners such as Minh Khang Center, increasing wide-scale deployment capability without investing in a traditional distribution network.

Promotion

For a B2B technology product like Liv5-IRS, the survey shows that customers typically do not make decisions based on mass advertising. Instead, they value real-world evidence of the system's operability, input from industry experts, and direct experience before deciding to cooperate.

This was clearly demonstrated during the product demonstration at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026). After directly experiencing the system, the leadership of the Vietnam Basketball Federation (VBF), the Head of the VBF Referees Committee, the referee team, organizers, and sponsors all recognized the practicality and application potential of Liv5-IRS at tournaments in Vietnam. The event also opened many cooperation opportunities with partners in the sports field and became an important proof point of the product's value.

From this insight, the team determined that Liv5-IRS's communication strategy will focus on real demonstrations, pilot deployments at tournaments, building successful case studies, and leveraging the industry partner network, rather than investing heavily in mass advertising. For the target group of organizers and tournament operators, real-world evidence and referrals from reputable partners are considerably more persuasive than traditional advertising.

2.2.7. Product Validation Through Real Deployment (Prototype Validation)

After completing market research and building the first version of Liv5-IRS, the team did not stop at gathering input through surveys but proceeded to test the product in a real competition environment to assess the system's operability, stability, and suitability for match officiating.

The prototype of Liv5-IRS was successfully demonstrated at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026). This was the first time the product was brought into a real competition

environment, with the participation of referees, organizers, representatives of the Vietnam Basketball Federation (VBF), and many tournament sponsors.

Unlike surveys or demonstrations in a meeting room, testing at an official tournament gave the team a more comprehensive assessment of the system's operation under real conditions, including video processing speed, stability when working with multiple camera feeds, the ability to support referees during situation review, and the user experience.

During the demonstration, the system operated stably and met the core functions set out, including multi-angle replay, slow motion, frame-by-frame viewing, and intuitive control through a referee-dedicated interface. Thanks to the optimized operating workflow, referees could become familiar with the system after only about five minutes of instruction, significantly reducing pre-tournament training time.



Figure 2.3. *Liv5-IRS demonstrated at the NYCC 3x3 2026 tournament.*

Feedback from Stakeholders

After the real-world experience, the team gathered feedback from various stakeholders to assess the product's suitability.

The leadership of the Vietnam Basketball Federation (VBF) commended the direction of developing an Instant Replay solution suited to the conditions of domestic tournament organization, especially in the student and semi-professional segment.

The referee team and the Head of the VBF Referees Committee recognized that Liv5-IRS's interface is intuitive, easy to approach, and suited to referees' workflow. Actions such as switching angles, slow motion, or frame-by-frame viewing were all performed quickly, helping shorten the time to handle situations requiring review.

In addition, sponsors attending the tournament expressed interest in the product and a desire to explore cooperation opportunities on future projects. This shows that Liv5-IRS attracts interest not only from direct users but also from businesses operating in the sports field.

Lessons Learned from the Testing Process

Alongside the positive feedback, the real-world deployment also helped the team identify points to continue improving before commercializing the product.

First, the processing capability of the AI modules depends significantly on hardware configuration. On computers equipped with high-performance GPUs, the system can fully exploit features such as AI Tracking, AI Image Enhancement, and AI Assistant. However, on general-purpose devices, activating all AI modules can affect processing performance.

From this result, the team decided to develop two product tiers. The standard version focuses on the core functions of an Instant Replay System and integrates only basic AI features to ensure operability on most devices. The advanced version, meanwhile, fully exploits the AI modules on high-performance hardware.

Second, the team found that the need for user training was lower than expected. Thanks to an interface designed around the referee's workflow, users could become familiar with and proficient in the basic operations after only about five minutes of instruction. This is a significant advantage over systems with complex interfaces or those requiring a dedicated technical team.

Finally, the team found that cooperating with units already providing livestreaming services or hardware solutions is more effective than deploying the entire system themselves. This led to the decision to cooperate with Minh Khang Center to integrate Liv5-IRS into an Instant Replay solution package and expand customer reach through the partner network.

Significance for Product Development

The deployment at NYCC 3x3 2026 marked an important shift for Liv5-IRS from the idea-development stage to product validation in a real environment. Unlike relying only on surveys or customer interviews, operating a pilot at an official tournament helped the team confirm that the product can meet real user needs and has commercialization potential.

The feedback gathered from the testing process was also used to adjust the user interface, guide the development of the AI modules, refine the business model, and build a partner cooperation strategy for the next phase.

2.3. Value Proposition Canvas

After market research, customer surveys, and product testing, the team reviewed the Value Proposition Canvas built in the early stage of the project. The results show that the initial customer assumptions still fit real-world needs; however, some components were supplemented based on survey results and the product demonstration at NYCC 3x3 2026.

The Value Proposition Canvas below reflects the complete version of Liv5-IRS after multiple rounds of validation with customers and partners.

Customer Profile	Value Map
Customer Jobs - Organize professional and transparent basketball tournaments. - Make fast and accurate referee decisions. - Expand value-added livestream services.	Products & Services - Instant Replay System (IRS). - Multi-camera replay (up to 16 cameras). - Slow Motion and Frame-by-frame. - AI Tracking, AI Image Enhancement, AI Assistant. - Referee-friendly interface. - Flexible deployment on existing hardware.
Customer Pains - High cost of professional replay systems. - Existing livestream software is difficult for referees. - Disputed decisions without video evidence. - Time-consuming training and technical setup.	Pain Relievers - Low-cost deployment using existing infrastructure. - Simple workflow with minimal training. - AI assists referees without replacing their decisions. - Compatible with various hardware configurations.
Customer Gains - Faster and more accurate decisions. - Fairer and more transparent tournaments. - Higher organizer credibility. - New revenue opportunities for livestream providers.	Gain Creators - Improves tournament professionalism. - Reduces review time and referee pressure. - Cost-effective alternative to expensive IRS solutions. - Validated through real-world deployment at NYCC 3x3 2026.

Table 2.2. *Liv5-IRS Value Proposition Canvas*

2.3.1. Customer Profile

Customer Jobs

Liv5-IRS's customers comprise three main groups: tournament organizers, the referee team, and livestreaming providers. Although each group has a different role, all share the common goal of raising the quality and professionalism of the tournament.

For organizers, the most important job is to ensure the tournament is fair, minimize disputes, and build a professional image to attract teams, sponsors, and spectators. Applying technology to support referees also helps raise the tournament's reputation over the long term.

For referees, the main job is to make accurate decisions within a short time, especially in fast-moving situations or those that directly affect the outcome. They therefore need a system that can quickly retrieve angles, is simple to operate, and does not interrupt the pace of play.



For livestreaming providers, the goal is to expand their service portfolio and create added value for customers. Integrating Liv5-IRS lets them not only provide broadcasting services but also deploy an Instant Replay solution for tournament organizers.

Customer Pains

Through surveys and direct discussions, the team identified several key difficulties customers face.

First, the investment cost of professional Instant Replay systems is too high relative to the budgets of most tournaments in Vietnam.

Second, common livestreaming software is not designed specifically for referees, leading to complex operation and heavy dependence on technicians.

Third, many tournaments still make decisions based only on direct observation, making it hard to handle controversial situations objectively.

In addition, training on and operating existing systems often takes considerable time and requires a dedicated technical team, increasing organizing costs.

Customer Gains

Customers want a system that helps referees make more accurate decisions while remaining fast and easy to use.

Organizers expect the solution to improve officiating quality and build trust with teams and sponsors without significantly increasing organizing costs.

Livestreaming providers, meanwhile, want to expand their services, increase competitiveness, and make effective use of their existing camera infrastructure.

2.3.2. Value Map

Products & Services

Liv5-IRS is an Instant Replay System developed specifically for basketball tournaments in Vietnam. Built with a software-first approach, it can be deployed on existing camera and computer systems or combined with equipment-providing partners.

The system's main functions include:

- Multi-angle video review.
- Synchronization of up to 16 cameras.
- Slow motion.
- Frame-by-frame viewing.
- Bookmarking and fast retrieval of situations.
- AI Tracking to support player tracking.
- AI Image Enhancement to improve image quality.
- AI Assistant to support search and operation.
- A referee-optimized interface.
- Flexible deployment across many hardware configurations.

**Pain Relievers**

Liv5-IRS is designed to directly address the difficulties customers face.

Rather than requiring investment in an expensive dedicated hardware system, Liv5-IRS leverages existing infrastructure or partners to significantly reduce deployment cost.

The system's interface is built around the referee's workflow, so users can become familiar with and proficient in it after only about five minutes of instruction.

The AI features are developed to support situation analysis, enhance image quality, and streamline operation, but do not replace the referee's decision-making authority. This ensures transparency and aligns with FIBA rules.

In addition, the system has an open architecture, allowing deployment on many computer configurations. On devices with high-performance GPUs, users can fully exploit the AI features; on general-purpose computers, users can still access all the core functions of the Instant Replay System.

Gain Creators

Liv5-IRS delivers value to customers by combining flexible deployment, reasonable cost, and modern technology.

For organizers, the system helps improve transparency in match officiating, reduce disputes, and increase professionalism, thereby raising reputation with teams, spectators, and sponsors.

For referees, Liv5-IRS helps shorten the time to handle situations requiring review, reduces pressure during decision-making, and improves the user experience through an intuitive interface.

For livestreaming providers, integrating Liv5-IRS creates a new value-added service, helping expand the product portfolio and increase market competitiveness.

In addition, the successful deployment at NYCC 3x3 2026, along with positive feedback from VBF and partners, helps reinforce customer confidence in the product's feasibility.

Chapter 3. Business Model

After identifying market needs, analyzing competitors, and validating the problem through customer surveys, the team built a business model for Liv5-IRS to define how to create, deliver, and capture value from the product. The business model is developed based on the Business Model Canvas (BMC) of Osterwalder & Pigneur (2010), adapted to the characteristics of the sports-technology market in Vietnam and the project's long-term development direction.

Liv5-IRS is positioned as a Business-to-Business (B2B) solution, providing an Instant Replay System (IRS) for basketball tournaments through software deployment combined with technical support services. Rather than selling the software as a perpetual license, the project aims for a per-tournament service model to reduce customers' initial investment and create recurring revenue for the business.

Through this model, Liv5-IRS provides not only referee-support software but also a complete solution including deployment consulting, usage training, technical support, and product improvement based on real customer feedback.

3.1. Business Model Canvas

The Business Model Canvas (BMC) is a tool that describes how a business creates, delivers, and captures value through nine core components. For Liv5-IRS, the BMC is built based on customer surveys, competitor analysis, and real-world testing at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026).

Unlike professional Instant Replay systems with large investment costs, Liv5-IRS focuses on providing a flexible, easy-to-deploy solution suited to the organizing conditions of tournaments in Vietnam. The entire business model is therefore designed around the goal of lowering the barrier to accessing technology while ensuring effectiveness in match officiating.

Component	Content
Customer Segments	Tournament organizers; livestreaming providers; federations and sports centers; recording-equipment providers.
Value Proposition	Affordable Instant Replay system; support for up to 16 cameras; AI to support referees; fast deployment; easy to use; no need to invest in dedicated hardware.
Channels	Direct sales; livestreaming partners; Minh Khang Center partnership; website; social media; demonstrations at tournaments; Vietnam Basketball Federation partnership.
Customer Relationships	Pre-tournament consulting; usage training; technical support; maintenance and updates; post-tournament feedback collection.
Revenue Streams	Deployment fees for livestreaming providers; direct deployment fees for organizers; technical support services; future extended services.
Key Resources	The Liv5-IRS software platform; AI algorithms; the development team; data from tournaments; deployment experience; the partner network.

Key Activities	Software development; AI research; deployment at tournaments; customer support; feedback collection and product improvement.
Key Partners	Minh Khang Center; livestreaming providers; tournament organizers; Vietnam Basketball Federation (VBF); recording-equipment suppliers.
Cost Structure	Staffing costs; R&D costs; AI costs; deployment costs; operating costs; marketing and customer care.

Table 3.1. *Liv5-IRS Business Model Canvas*

Business Model Canvas shows that Liv5-IRS focuses not only on product development but also on building an ecosystem of cooperation with partners in the sports industry to expand deployment capability and accelerate commercialization. Combining a technology platform, deployment services, and a partner network helps the project create a sustainable competitive advantage in a Vietnamese Instant Replay market that is still in its early stages.

3.2. Customer Interface

The Customer Interface describes how Liv5-IRS reaches customers, delivers value, and builds lasting relationships with each target segment. For a B2B technology product like Liv5-IRS, the success of the business model depends not only on software quality but also on the ability to deploy, train, provide technical support, and maintain customer relationships after each tournament.

Unlike ordinary software products that can be distributed online and used immediately after purchase, Liv5-IRS is a solution that directly serves match officiating. The customer-acquisition process is therefore always tied to consulting, site assessment, system deployment, user training, and operating support throughout the tournament. This means the business sells not just software but a complete solution that meets each customer's real needs.

3.2.1. Customer Segments

Based on market research, needs analysis, and real-world testing at NYCC 3x3 2026, Liv5-IRS identifies four main customer groups.

Tournament organizers

This is Liv5-IRS's most important customer segment, as they are the party that directly decides whether to invest in or rent an Instant Replay service. Organizers are responsible for the professional quality, transparency, and image of the tournament, while also having to balance the organizing budget.

Through the survey, the team found that most organizers want to apply technology to support referees but face a barrier in the cost of investing in professional Instant Replay systems. Liv5-IRS is therefore positioned as an affordable, easy-to-deploy solution that can leverage existing infrastructure rather than requiring an entirely new system investment.

For this customer group, the value Liv5-IRS delivers is not only the ability to review situations but also reducing disputes, increasing fairness in play, and raising the professionalism of the tournament.



Livestreaming and image-production providers

Livestreaming providers serve as both a customer group and a strategic partner for Liv5-IRS. They own camera systems, recording equipment, and technical staff, so integrating Liv5-IRS into their existing workflow significantly reduces deployment costs for both sides.

Through this form of cooperation, livestreaming providers can expand the service portfolio they offer customers, while Liv5-IRS gains the opportunity to reach more tournaments without building a nationwide deployment team.

Liv5-IRS has built a cooperation model with livestreaming providers based on per-tournament software licensing at a price of VND 5–10 million, depending on scale and deployment requirements.

Federations, sports centers, and tournament governing bodies

This customer group plays an important role in driving the adoption of referee-support technology in Vietnam. Unlike organizers who manage a single specific tournament, federations and sports centers typically organize many tournaments per year and have significant influence over competition-organizing standards.

The fact that Liv5-IRS was demonstrated at NYCC 3x3 2026 and received positive feedback from the leadership of the Vietnam Basketball Federation (VBF) and the Head of the VBF Referees Committee is a positive signal that the product can meet professional requirements and has the potential to be deployed more widely in official tournaments in the future.

Equipment and technology-solution partners

Beyond direct customers, Liv5-IRS also builds cooperative relationships with businesses providing recording equipment and sports-technology solutions to expand deployment capability.

Liv5-IRS is currently working with Minh Khang Center to integrate the software into its Instant Replay solution package. This cooperation leverages the partner's customer network, deployment experience, and existing technical infrastructure, while reducing the cost of market expansion in the early phase.

In the future, Liv5-IRS will continue to expand cooperation with businesses providing cameras, recording equipment, dedicated computers, and sports event organizers to build a comprehensive support ecosystem.

3.2.2. Channels

Because Liv5-IRS is a high-value B2B solution requiring direct deployment at the tournament venue, the project does not choose a mass-distribution model but instead focuses on channels that build trust and create real-world experience for customers.

Direct Sales

This is the most important channel in the current phase. The Liv5Solutions team proactively contacts tournament organizers, livestreaming providers, and sports centers to introduce the product, demonstrate the system, and advise on suitable deployment options.

Direct contact helps the team clearly understand each customer's needs, assess the existing infrastructure, and propose a solution suited to the scale of each tournament.

Cooperation with strategic partners

Liv5-IRS prioritizes market expansion through partners already experienced in tournament organization and sports livestreaming.



The cooperation model with Minh Khang Center is a prime example of this strategy. Through the partner, Liv5-IRS can reach more potential customers, reduce sales costs, and leverage a technical team already experienced in tournament deployment.

In the future, the team will continue to expand cooperation with sports-technology businesses, recording-equipment providers, and event organizers to increase the product's market coverage.

Field Demonstration

For technology products that serve referees, real-world experience matters far more than traditional advertising.

Liv5-IRS therefore chooses a strategy of pilot deployment at tournaments so that customers can directly observe the operating workflow and assess the system's effectiveness.

The successful demonstration at NYCC 3x3 2026 not only helped verify the product's feasibility but also helped build brand reputation and open many cooperation opportunities with units in the sports industry.

Digital media channels

Alongside direct sales, Liv5-IRS also builds media channels to introduce the product and maintain brand presence in the digital environment. These channels include the official website, a Facebook fan page, YouTube, and LinkedIn, together with product brochures, demo videos, and case studies from tournaments already deployed.

Although this is not the main revenue channel, it plays an important role in providing information, building reputation, and supporting customers as they research the product before deciding to cooperate.

3.2.3. Customer Relationships

Liv5-IRS aims to build lasting customer relationships rather than merely providing software for each tournament. The entire interaction is designed as an accompaniment model — before, during, and after deployment — to ensure the system operates stably and is continuously improved based on real feedback.

Before the tournament

The Liv5Solutions team works directly with the customer to survey the infrastructure, determine the number of cameras, assess technical conditions, and advise on a suitable deployment option. After agreeing on the plan, the team installs the system, configures the software, and trains referees and the operating team.

During the tournament

Throughout the tournament, the Liv5Solutions technical team continuously monitors and supports operations to ensure the system runs stably. If equipment or software issues arise, the team coordinates with the organizer and livestreaming provider to resolve them promptly, minimizing impact on match officiating.

After the tournament

After the tournament ends, Liv5Solutions continues to maintain the customer relationship by collecting feedback, assessing the user experience, and recording improvement suggestions. This input is compiled to prioritize feature development in subsequent versions.

This cooperation model helps Liv5-IRS continuously refine the product based on real market needs, while building trust and increasing the likelihood that customers continue to use the service in future tournaments.

3.3. Infrastructure Management

Infrastructure Management describes how Liv5-IRS organizes and operates resources to create value for customers. For a B2B technology product like Liv5-IRS, the business's value lies not only in the software but also in its research, deployment, technical support, and partner network in the sports field.

Unlike businesses selling physical products, Liv5-IRS operates a model that combines technology development with service provision. The three components — Key Activities, Key Resources, and Key Partners — are therefore closely linked, helping ensure the product is continuously improved, deployed effectively, and able to meet real customer needs.

3.3.1. Key Activities

To create and sustain customer value, Liv5-IRS focuses on five core activity groups: product development, AI technology research, system deployment at tournaments, customer support, and product improvement based on real feedback.

Developing and refining the product

Liv5-IRS's most important activity is researching, developing, and maintaining the Instant Replay System platform to meet the increasing needs of basketball tournaments. The development team continuously optimizes the user interface, improves video processing performance, enhances system stability, and expands compatibility with various recording devices.

During development, the team always prioritizes features that deliver direct value to referees — such as instant replay, slow motion, frame-by-frame viewing, situation bookmarking, and multi-angle management — rather than features oriented toward television program production.

AI research and development

One of Liv5-IRS's long-term directions is applying artificial intelligence to improve referee support while ensuring the Human-in-the-loop principle, in which AI plays only a supporting role and the referee makes the final decision.

AI research currently focuses on three main modules: AI Tracking, AI Image Enhancement, and AI Assistant. These modules are developed to support player tracking, improve image quality in hard-to-observe situations, and help users operate faster during review.

Besides developing the algorithms, the team also emphasizes optimizing operability across many hardware configurations. On computers with high-performance GPUs, the system fully supports the AI features; on general-purpose devices, the system still ensures all core Instant Replay functions and only limits certain AI features that require large processing resources.

System deployment at tournaments

Unlike software that can be installed and used entirely online, Liv5-IRS must be deployed directly at the tournament venue. The deployment process therefore plays an important role in ensuring the system operates stably and meets the customer's requirements.

For each tournament, the Liv5Solutions team surveys the venue, assesses the technical infrastructure, configures the system, synchronizes cameras, and tests the entire workflow before the tournament



officially begins. During the competition, the technical team continues to monitor the system and stands ready to resolve any issues to ensure Instant Replay runs continuously.

The real-world deployment at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026) demonstrates Liv5-IRS's operability under real competition conditions, while helping the team verify the product's stability and gather valuable user feedback.

Customer training and support

Beyond developing the technology, Liv5-IRS places particular emphasis on the user experience. After deployment is complete, the Liv5Solutions team directly guides referees and the operating team in using the system.

Thanks to an intuitive interface optimized around the referee's workflow, users can become familiar with and proficient in the system after only about five minutes of training. This significantly reduces pre-tournament preparation time and limits errors during operation.

Throughout usage, customers also receive technical support from the development team to ensure the system operates stably and any arising problems are resolved promptly.

Feedback collection and product improvement

Liv5-IRS applies a Continuous Improvement approach to product development, in which each tournament is viewed as an opportunity to assess the system's effectiveness and identify new customer needs.

After each deployment, the team collects feedback from referees, organizers, livestreaming providers, and technical partners to assess satisfaction and record improvement suggestions. This feedback becomes an important basis for prioritizing feature development and raising product quality in subsequent versions.

3.3.2. Key Resources

To operate and develop Liv5-IRS, the project combines resources across technology, people, and a cooperation network. These resources not only help create the product but also play an important role in maintaining competitive advantage and expanding the business in the future.

The Liv5-IRS software platform

The project's most important resource is the Instant Replay software platform that the team researched and developed in-house. The system is designed specifically for basketball refereeing, with the ability to synchronize up to 16 cameras and support instant replay, slow motion, frame-by-frame viewing, situation bookmarking, replay-video export, and real-time multi-angle management.

Owning a proprietary technology platform lets Liv5-IRS take the initiative in upgrading, adding features, and optimizing the system to meet customer needs without depending on foreign providers.

AI technology and data

Another important resource is the AI modules and the dataset built during research and real-world deployment. Data collected from tournaments helps the team continue training and improving the AI algorithms, thereby raising the accuracy of AI Tracking, AI Image Enhancement, and AI Assistant.

Combining AI technology with real-world data creates a lasting competitive advantage, because this is an asset that is hard to copy and grows more valuable as the number of deployed tournaments increases.

**The development team**

The Liv5Solutions team is the core resource determining the project's success. It comprises members with knowledge of software development, image processing, artificial intelligence, and multimedia technology.

Beyond technical capability, the team has accumulated substantial experience by researching customer needs, participating in tournament deployments, and working directly with referees, organizers, and livestreaming providers. Combining technology knowledge with tournament-organizing practice helps the team develop a product closely aligned with market needs.

Deployment experience and brand reputation

Unlike many projects that stop at the idea or simulation stage, Liv5-IRS has been validated through real-world deployment at NYCC 3x3 2026. This is an intangible but highly valuable resource that helps raise the project's reputation and build trust with potential customers.

In addition, receiving positive feedback from the leadership of the Vietnam Basketball Federation (VBF), the Head of the VBF Referees Committee, and industry partners helps Liv5-IRS build an image as a technology solution with real application capability, rather than merely a pilot product.

The partner network

Beyond internal resources, Liv5-IRS also has a cooperation network with units operating in the sports field. Relationships with Minh Khang Center, livestreaming providers, tournament organizers, and referee experts help the project expand deployment capability, reach new customers, and continuously update market needs.

This partner network supports not only business operations but also plays an important role in validating the product, improving service quality, and driving commercialization.

3.3.3. Key Partners

For a B2B technology project like Liv5-IRS, building a strategic partner network is no less important than product development. Rather than performing all deployment and business activities itself, Liv5-IRS chooses a cooperation model to leverage the resources, experience, and customer networks of units in the sports industry.

Minh Khang Center

Minh Khang Center is currently Liv5-IRS's strategic partner in integrating the software into the unit's Instant Replay solution package. This cooperation helps Liv5-IRS expand customer reach, leverage the partner's real-world deployment experience, and reduce infrastructure investment costs in the early development phase.

At the same time, combining Liv5-IRS's software platform with Minh Khang Center's equipment systems helps create a complete solution that better meets the needs of tournaments of varying scale.

Livestreaming providers

Livestreaming providers are both customers and deployment partners for Liv5-IRS. They own camera systems, recording equipment, and technical staff, so they can integrate Liv5-IRS into their existing production workflow without much additional investment.

This cooperation model benefits both sides: Liv5-IRS expands the market faster, while livestreaming providers gain an additional value-added service to offer customers.



Organizers and the Vietnam Basketball Federation

Tournament organizers and the Vietnam Basketball Federation (VBF) are important partners in validating and refining the product. Professional feedback from the referee team, organizers, and VBF leadership helps Liv5-IRS better understand the real requirements of match officiating, thereby prioritizing the development of appropriate features.

Although VBF is not yet a commercial partner, the positive recognition from experts and leaders in the industry is an important basis for the project to build reputation and expand cooperation opportunities in the future.

Recording-equipment and technology-infrastructure suppliers

To ensure deployment capability across tournaments of varying scale, Liv5-IRS maintains cooperative relationships with providers of cameras, computers, and recording equipment. Leveraging partners' existing infrastructure reduces initial investment costs while increasing flexibility when deploying at many different venues.

The role of the partner network in the business model

The partner network helps Liv5-IRS shorten time-to-market, reduce deployment costs, and improve future scalability. Rather than investing all resources in building its own ecosystem, the project chooses a model of cooperating with units that have expertise and existing customer networks. This is a strategy well-suited to an early-stage technology startup, helping optimize resources and build a foundation for sustainable commercialization.

Chapter 4. Market Entry Strategy

After completing market research, validating customer needs, and testing the product in a real environment, Liv5-IRS enters the phase of building a Market Entry Strategy. The goal of this strategy is to bring the product to the right customer segment, optimize the team's resources in the initial phase, and gradually build a position in the sports-technology market.

Because Liv5-IRS is a B2B technology solution with customers mainly comprising tournament organizers, livestreaming providers, and sports organizations, the market-entry strategy does not focus on rapidly expanding the number of individual users but instead prioritizes building reputation through real-world deployments, forming a partner network, and developing lasting customer relationships.

4.1. Product Strategy

Liv5-IRS is positioned as an Instant Replay System (IRS) for basketball tournaments in Vietnam, aiming to become the technology solution that supports referees during match officiating. Unlike professional Instant Replay systems worldwide, which require large investment, dedicated infrastructure, and a highly skilled operating team, Liv5-IRS is developed with a direction of being affordable, easy to deploy, and suited to the organizing conditions of tournaments in Vietnam.

Rather than competing directly with high-end systems in the international segment, Liv5-IRS chooses a strategy of focusing on the gap in the domestic market — where many student, semi-professional, and local tournaments need to apply technology but cannot yet afford traditional Instant Replay solutions. This lets Liv5-IRS create a competitive advantage through wider reach, fast deployment time, and cost suited to customers' budgets.

One of Liv5-IRS's core values is that AI plays a supporting role only and does not replace the referee. The AI modules are designed to support player tracking, improve image quality, and streamline operation, while the final decision remains with the referee, fully in keeping with the spirit of the FIBA Basketball Rules. This direction lets the product leverage the advantages of artificial intelligence while ensuring transparency and alignment with the current match-officiating workflow.

Beyond technology, Liv5-IRS also emphasizes the user experience. Through real-world testing at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026), the team found that the intuitive interface and simple operating workflow are among the product's standout advantages. Referees can become familiar with and proficient in the system after only about five minutes of training, helping shorten pre-tournament preparation time and reduce pressure on the operating team.

Current feature set

As of the completion of this report, Liv5-IRS has finalized the core functions of an Instant Replay system for refereeing, including:

- Support for synchronizing up to 16 cameras operating simultaneously.
- Reviewing situations from multiple camera angles.
- Instant replay while the match is still ongoing, ensuring the system continues recording continuously and does not miss subsequent situations.
- Slow motion.
- Zoom to observe situation details.
- Frame-by-frame viewing.
- Bookmarking and fast retrieval of important situations.



- Exporting replay clips as separate files for storage or sharing.
- A control interface optimized for referees.
- Stable operation across many computer configurations.

Beyond the core Instant Replay functions, Liv5-IRS has researched and continues to develop AI modules to improve effectiveness, including:

- AI Tracking to support player tracking within the frame.
- AI Image Enhancement to improve image quality in situations where the recording quality is poor.
- AI Assistant to help users search for situations, operate quickly, and streamline system usage.

To accommodate varied deployment conditions, Liv5-IRS is designed with a Scalable Architecture. On computers with high-performance GPUs, users can access all AI features. On general-purpose devices, the system still ensures all essential Instant Replay functions while automatically optimizing or limiting certain AI features that require large processing resources to maintain stable performance.

Product development direction

For the period 2026–2031, Liv5-IRS aims to develop under a One Platform, Multiple Solutions strategy. Rather than developing many separate products, the team focuses on refining a single unified Instant Replay platform and gradually expanding its applications through new functional modules.

In the short term, the project will continue to refine the AI modules to improve referee support, optimize image quality, and enhance the user experience, while maintaining the principle of “AI supports — the referee decides.”

In the medium term, Liv5-IRS will expand compatibility with various cameras, recording devices, and technical infrastructure to reduce deployment cost, increase flexibility, and allow the product to be applied to tournaments of many scales.

In the long term, Liv5-IRS aims to develop into a technology platform supporting sports officiating, integrating functions for video management, data storage, match statistics, and analysis, while researching expansion into other sports such as volleyball, futsal, and badminton. This strategy not only increases value for existing customers but also expands the market size and builds a foundation for the project's sustainable development in the future.

4.2. Pricing Strategy

Liv5-IRS applies a Value-based Pricing strategy, in which the price is built on the value the product delivers to customers rather than solely on software development cost.

Through surveys and discussions with tournament organizers, the team found that most customers do not want to invest in a high-value Instant Replay system but instead prefer to rent the service per tournament. Liv5-IRS therefore chooses a per-project pricing model to lower the initial investment barrier and increase customer access.

The project currently offers two forms of cooperation.

For livestreaming providers, Liv5-IRS offers software usage rights combined with technical support at an expected price of VND 5–10 million per tournament lasting about four weeks. The fee varies according to tournament scale, the number of cameras, and the scope of support.



For tournament organizers, Liv5-IRS offers an all-inclusive deployment package at a starting price of VND 5 million per tournament, with the specific cost negotiated based on actual requirements.

This pricing strategy helps Liv5-IRS create a competitive advantage in the student and semi-professional tournament segment, while enabling customers to experience the product before considering long-term cooperation.

4.3. Distribution (Place) Strategy

As a B2B technology solution, Liv5-IRS is not distributed through retail channels but instead focuses on direct sales combined with a partner network in the sports field.

In the initial phase, the team prioritizes reaching tournament organizers directly through product demonstrations, solution introductions, and consulting on deployment options suited to each tournament. This approach gives customers the chance to experience the product firsthand while enabling the team to gather feedback and refine the product.

In parallel, Liv5-IRS expands market access through partners operating in livestreaming and sports-equipment provision. Liv5-IRS is currently working with Minh Khang Center to integrate the software into the partner's Instant Replay solution package. This cooperation leverages Minh Khang Center's customer network, deployment experience, and existing infrastructure, thereby shortening time-to-market.

In the future, the project will continue to expand cooperation with livestreaming providers, recording-equipment suppliers, and sports organizations to build a nationwide deployment ecosystem.

For each customer, Liv5-IRS applies a standardized deployment process comprising needs assessment, infrastructure evaluation, system installation, referee training, technical support during the tournament, and post-tournament feedback collection. This process helps ensure consistent service quality and maintain lasting customer relationships.

4.4. Promotion Strategy

Given that the product is a specialized technology solution for sports organizations, Liv5-IRS prioritizes a communication strategy focused on building reputation and trust through real-world evidence rather than mass promotion.

For Owned Media, the team uses the website, a Facebook fan page, LinkedIn, and product brochures to provide official information about Liv5-IRS. Demo videos, real-world deployment images, and user guides are also produced to help customers better understand the product before deciding to cooperate.

For Earned Media, Liv5-IRS focuses on effectively leveraging real-world deployment activities. The successful product demonstration at NYCC 3x3 2026, along with positive feedback from the leadership of the Vietnam Basketball Federation (VBF), the Head of the VBF Referees Committee, and sponsors, has helped raise the product's credibility. In the next phase, case studies, customer feedback, and referrals from partners will continue to be important communication channels.

For Paid Media, the project uses only a reasonable budget to support reaching the right customers. Preferred forms include Facebook advertising aimed at the basketball community, Google Ads for keywords related to refereeing technology, and sponsoring industry seminars or events.



Alongside communication activities, Liv5-IRS also runs several sales-promotion programs, such as free demonstration sessions at tournaments, trial versions for livestreaming partners, and preferential policies for early customers. These activities let customers experience the product firsthand, thereby increasing conversion from the interest stage to formal cooperation.

Chapter 5. Risk Analysis and Development Direction

Commercializing a technology product in the sports field depends not only on technical quality but also on many factors from the market, finance, partners, and the legal environment. Although Liv5-IRS has initially demonstrated its feasibility through real-world testing at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026) and received positive feedback from the leadership of the Vietnam Basketball Federation (VBF), the Head of the VBF Referees Committee, and industry partners, the project is still in the early stages of commercialization and faces many challenges.

Therefore, identifying key risks, assessing their level of impact, and building appropriate response plans will help Liv5-IRS be more proactive during development, while improving its scalability and sustainable growth in the future.

5.1. Risk Analysis

To assess the project's risk level, the team uses two criteria — Impact and Likelihood — on a scale of 1 to 5. The assessment results are compiled based on market research, customer surveys, and experience from pilot deployment of the product.

Market risk

The Instant Replay market in Vietnam is still quite new, while most basketball tournaments today are small to medium in scale with limited organizing budgets. Although demand for greater transparency and higher officiating quality is rising, many organizers still prioritize their budget for the venue, prizes, media, or staff before investing in supporting technology solutions.

This could make Liv5-IRS's market expansion slower than expected, especially in the early commercialization phase.

Criterion	Rating
Impact	5/5
Likelihood	4/5

To mitigate this risk, Liv5-IRS chooses a per-tournament rental model rather than selling the software as a perpetual license. At the same time, the team prioritizes deploying at large-scale or influential tournaments to create real-world case studies, then gradually expanding to smaller tournaments. Cooperating with livestreaming providers also helps customers leverage existing infrastructure and significantly reduce deployment costs.

Competitive risk

Currently, the number of Instant Replay solutions dedicated to the Vietnamese basketball market is quite limited. However, as demand for technology in sport rises, the market may attract additional domestic businesses or international providers.

In addition, competitors with strong financial resources are fully capable of developing similar solutions or bringing existing products to the Vietnamese market.

Criterion	Rating
Impact	4/5
Likelihood	3/5

To maintain its competitive advantage, Liv5-IRS aims to continuously improve the product, refine the AI modules, build the partner network, and expand the number of deployed tournaments in the early phase to increase brand recognition and create a barrier to market entry.

Technology risk

As a real-time video processing system, Liv5-IRS requires high stability throughout operation. Issues such as camera signal loss, software errors, connection interruptions, or hardware configurations that do not meet requirements can directly affect referees' work during a match.

In addition, AI features such as AI Tracking or AI Image Enhancement require large processing resources, so performance can vary between devices.

Criterion	Rating
Impact	5/5
Likelihood	3/5

To mitigate this risk, the team builds a testing process before each tournament, designs backup mechanisms for signal-loss situations, and optimizes the software for various configuration levels. On low-specification devices, the system still ensures full core Instant Replay functions, while certain advanced AI features are adjusted to maintain stable performance.

Financial risk

In the early development phase, Liv5-IRS needs to sustain resources for research, product development, testing, and customer support while revenue is not yet stable.

If the number of real-world deployed tournaments is lower than planned, or if market expansion is slower than expected, the project may face pressure on cash flow and the ability to reinvest.

Criterion	Rating
Impact	4/5
Likelihood	3/5

To mitigate this risk, the team aims to expand cooperation with strategic partners such as livestreaming providers and equipment suppliers, while diversifying revenue through deployment packages, technical support services, and extended AI modules. Leveraging startup support programs and controlling operating costs are also important solutions in the early phase.

Legal and data-security risk

Liv5-IRS processes and stores image data during tournaments, so it must ensure compliance with data-protection regulations and agreements with organizers.

In addition, applying artificial intelligence to support referees must ensure transparency and must not change the decision-making role of humans.

Criterion	Rating
Impact	3/5
Likelihood	2/5

The team plans to build a clear data-management policy, storing data only with customer consent and complying with applicable laws. At the same time, all AI modules will be developed under the principle of “AI supports — the referee decides,” ensuring that every final decision is made by the referee.

Risk group	Impact	Likelihood	Priority level
Market	5	4	Very high
Technology	5	3	High
Competition	4	3	Medium
Financial	4	3	Medium
Legal and data	3	2	Low

Table 5.1. Risk assessment summary

5.2. Five-Year Development Plan

After completing market research, validating customer needs, and successfully deploying a pilot version, Liv5-IRS sets a five-year development roadmap to gradually refine the product, expand the market, and build a technology ecosystem supporting sports officiating.

Rather than expanding rapidly across many markets at once, the project chooses a phased development strategy, starting by consolidating the domestic basketball market before expanding into other sports and toward the regional market.

2026–2027: Refine the product and standardize the deployment process

In the first year after finalizing the MVP, Liv5-IRS focuses on optimizing system stability, improving video processing performance, and continuing to refine the AI modules. In parallel with product development, the team will continue deploying at more tournaments to gather user feedback, standardize the operating process, and build additional real-world evidence.

At the same time, the project will continue to expand cooperation with Minh Khang Center and technical partners to improve deployment capability across various infrastructure conditions.

2027–2028: Expand the domestic customer base

After the product reaches stability, Liv5-IRS will focus on expanding the customer network nationwide. The goal of this phase is to increase the number of tournaments using the system, expand cooperation

with livestreaming providers, and build a technical support team capable of deploying at many tournaments simultaneously.

In addition, the team will standardize referee training, user guides, and after-sales support services to improve the customer experience.

2028–2029: Build the product ecosystem

Beyond a single Instant Replay system, Liv5-IRS aims to develop into a platform supporting tournament officiating, with many integrated services such as video-data management, match statistics, situation storage and search, while expanding the AI features that support referees.

Forming a product ecosystem will help increase usage value and create additional revenue from extended services.

2029–2030: Diversify the market

After establishing its position in basketball, Liv5-IRS will research expanding the technology into other sports such as volleyball, futsal, badminton, or handball. Leveraging the existing technology platform reduces development cost and significantly expands the target market.

2030–2031: Toward the regional market

In the long term, Liv5-IRS aims to gradually commercialize the product in several Southeast Asian countries through cooperation with sports federations, technology partners, and event-organizing businesses. At the same time, the team will seek strategic investors to scale operations and develop an international version of the product.

Table 5.2. Liv5-IRS development roadmap, 2026–2031

Phase	Core focus
2026–2027	Refine the product, standardize the deployment process, and expand testing.
2027–2028	Expand customers and partners in the domestic market.
2028–2029	Build the product ecosystem and develop value-added services.
2029–2030	Expand the solution into other sports.
2030–2031	Reach the Southeast Asian market and seek strategic partners.



Prototype / MVP

Objectives of the Prototype/MVP

To verify the feasibility of the idea before commercialization, the team developed a Minimum Viable Product (MVP) of Liv5-IRS with the full core functions of an Instant Replay System (IRS) for basketball. The MVP was built to assess operability in a real competition environment, validate the needs of target customers, and gather user feedback to continue refining the product.

Unlike prototypes that stop at interface simulation or feature demonstration, Liv5-IRS's MVP is a system that can be deployed directly at a tournament, allowing simultaneous assessment of processing performance, user experience, integration with existing recording infrastructure, and stability during real operation.

Completed Functions

As of the completion of this report, Liv5-IRS's MVP has finalized the functions serving Instant Replay in basketball, including:

- Support for synchronizing up to 16 cameras operating simultaneously.
- Reviewing situations from multiple camera angles.
- Instant replay while the match is still ongoing — the system continues recording continuously in the background, so it does not interrupt or miss subsequent situations.
- Slow motion.
- Zoom to observe situation details.
- Exporting replay clips as separate files for storage or sharing after the match.
- A Bookmark system that marks important situations and retrieves them quickly at the exact recorded moment.
- Frame-by-frame viewing.
- An interface optimized for referees, with simple, intuitive controls.
- Performance optimization to run even on computers with medium or low configurations.
- Referees can become familiar with and proficient in the system after only about five minutes of training.

Beyond the core functions, the system has integrated and continues to develop AI modules to improve referee support, including:

- AI Tracking to support player tracking within the frame.
- AI Image Enhancement to improve image quality in situations where the recording quality is poor.
- AI Assistant to help users operate quickly, search for situations, and streamline system usage.

The AI features are developed under the principle of AI supports, does not replace the referee. The system does not make decisions automatically but only provides additional information and support tools so that referees can assess situations more accurately, consistent with the spirit of the FIBA Basketball Rules. The AI modules are also designed for multiple device tiers: on computers with high-performance GPUs, users can access all AI features; on general-purpose computers, the system still ensures all core Instant Replay functions and only limits certain AI features that require large processing resources to maintain stable performance.



Real-World Testing at NYCC 3x3 2026

After finalizing the MVP, Liv5-IRS was selected to be demonstrated and tested at the National Youth Clubs Basketball Cup — 3x3 2026 (NYCC 3x3 2026). This was an important milestone that helped the team assess the system's operability under real competition conditions and validate the product's suitability for target customers' needs.

During deployment, the system operated stably and met the requirements of Instant Replay work well. Referees could quickly become familiar with the interface and use the system proficiently after only about five minutes of instruction, showing that Liv5-IRS can be deployed in the real world without significantly changing the match-officiating workflow or increasing operating pressure on the referee team.

Beyond technical verification, the demonstration also helped the team gather many professional assessments from the leadership of the Vietnam Basketball Federation (VBF), the Head of the VBF Referees Committee, the referee team, tournament organizers, and sponsors. The feedback recognized the product's practicality, its application potential at domestic tournaments, and its potential to develop into an Instant Replay solution suited to tournament-organizing conditions in Vietnam.

Through this event, Liv5-IRS also expanded its cooperation network with partners in the sports field. In particular, the project began cooperating with Minh Khang Center to integrate Liv5-IRS into the unit's Instant Replay solution package. In addition, several partners and sponsors present at the tournament expressed a desire to continue discussing the possibility of deploying the product at future tournaments, creating a favorable foundation for commercialization.

Evidence of the MVP's validation in a real-world competition setting will be visually presented through a live demo during the in-class presentation. Alongside the demo, the team will introduce portfolio documentation containing real-world images, the system interface, the operating workflow, and feedback from referees, organizers, and partners.

Direction for Future Versions

After the real-world testing, the team identified several important directions for future versions of Liv5-IRS.

In the short term, the team will continue to optimize video processing performance, improve the user experience, enhance system stability, and refine the AI modules to support referees more effectively — while maintaining the principle that AI only supports, and the referee makes the final decision.

In the medium and long term, Liv5-IRS aims to expand compatibility with various recording devices and deployment infrastructure, integrate with tournament-management systems, and develop additional data-analysis and officiating-support features to build a comprehensive technology platform for the sports field. Once the product is refined and successfully commercialized in basketball, the team will also research expanding the technology into other sports that need an Instant Replay system, such as volleyball, futsal, and badminton.

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